



PRODUCT NAME

High-performance Toothpaste Formula

Description

Fluoride toothpaste with sorbitol, silica, fluoride, flavor, and preservative

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sorbitol 70%	Sorbitol	30%	Humectant, Sweetener
2	Hydrated Silica	Hydrated Silica	15%	Abrasive, Thickening Agent
3	Sodium Fluoride	Sodium Fluoride	0.24%	Anti-cavity Agent
4	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	1.5%	Surfactant
5	Flavor	Flavor	1%	Flavoring Agent
6	Sodium Saccharin	Sodium Saccharin	0.2%	Sweetener
7	Sodium Benzoate	Sodium Benzoate	0.5%	Preservative
8	Xanthan Gum	Xanthan Gum	0.6%	Thickening Agent
9	Purified Water	Aqua	49.96%	Solvent
10	Titanium Dioxide	Titanium Dioxide	1%	Opacifying Agent
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Phase 9: Manufacturing Phase

Phase 10: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions